EVision – Electric Vehicle Charging Demand Analytics

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

By

A Ramcharan Reddy :2420030147

P Chaitanya Kumar :2420030760

Y Durga Prasad Reddy :2420030281

Under the Esteemed Guidance of

Dr. Prathusha (Assistant Professor)

Designation

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Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

K L (Deemed to be) University
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING



Declaration

The Project Report entitled "**EVision – Electric Vehicle Charging Demand Analytics**" is a record of Bonafide work of **A. Ramcharan Reddy – 2420030147, P. Chaitanya Kumar – 2420030760, Y. Durga Prasad Reddy – 2420030281** submitted in partial fulfillment for the award of B. Tech in CSE to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

A. Ramcharan Reddy– 2420030147

P. Chaitanya Kumar – 2420030760

Y. Durga Prasad Reddy – 2420030281

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING



CERTIFICATE

This is to certify that the mini project based report entitled "**EVision – Electric Vehicle Charging Demand Analytics**" is a Bonafide work done and submitted by **A. Ramcharan Reddy – 2420030147, P. Chaitanya Kumar – 2420030760, Y. Durga Prasad Reddy – 2420030281** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of CSE, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Guide

Signature of the Course Coordinator **Signature of the HOD**

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A. Ramcharan Reddy– 2420030147

P. Chaitanya Kumar – 2420030760

Y. Durga Prasad Reddy – 2420030281

Abstract

The increasing adoption of electric vehicles (EVs) has created a growing need for reliable charging infrastructure and efficient electricity demand management. Hyderabad is experiencing growth in electric mobility, making it important to understand charging infrastructure availability, electricity consumption patterns, and future charging requirements. EVision – Electric Vehicle Charging Demand Analytics is a web-based application designed to support the analysis and visualization of electric vehicle charging infrastructure and demand in Hyderabad.

The application provides an interactive dashboard with features such as EV charging station discovery, station mapping, route planning, charging cost estimation, demand analytics, forecasting visualization, and infrastructure planning. It is developed using React.js, TypeScript, Vite, and Tailwind CSS to provide a responsive and user-friendly interface. Hyderabad-specific charging station information and monthly electricity consumption datasets are considered as the primary data sources for regional analysis. Historical consumption information can be used to identify demand trends and support future forecasting studies.

The main aim of EVision is to make EV charging information easier to understand and help users and infrastructure planners make informed decisions. The application provides a foundation for data-driven analysis of Hyderabad's EV charging ecosystem. Future enhancements can include real-time station availability, database integration, and machine learning-based demand prediction to improve the accuracy and usefulness of the system.

Index

S. No.	Chapters	Topics	Page.no
		Acknowledgement	
		Abstract	
1	Introduction	1.1 Background of the project 1.2 Problem statement 1.3 Scope of the project 1.4 Objective(s) 1.5 Importance of the application 1.6 Target users/audience	1-10
2	System Requirements	2.1 Hardware requirements 2.2 Software requirements 2.3 Development tools and frameworks	11-20
3	Technology Stack	3.1 Front-end (e.g., React.js, Angular, HTML/CSS) 3.2 Back-end (e.g., Node.js, Django, Spring Boot) 3.3 Database (e.g., MongoDB, MySQL, PostgreSQL) 3.4 Version Control (e.g., Git, GitHub) 3.5 APIs or External services (e.g., Firebase, Stripe)	21-30
4	System Architecture	4.1 High-level architecture diagram 4.2 Description of each layer (frontend, backend, database) 4.3 Deployment architecture (e.g., cloud, local, CI/CD pipelines)	31-35
5	Design	5.1 Wireframes or mockups 5.2 ER Diagram / Database schema	36-42
6	Implementation	6.1 Module-wise implementation 6.2 Front-end logic (UI rendering, state management) 6.3 Backend logic (routes, controllers, business logic) 6.4 Database connectivity 6.5 API integration 6.6 Authentication & Authorization	43-55
7	Features	7.1 List of main features 7.2 How each feature works (user perspective + technical description)	56-60
8	Testing	8.1 Postman testing (frontend/backend) 8.2 Integration testing 8.3 Tools used for testing 8.4 Test cases and results	61-67
9	Deployment	9.1 Steps to deploy 9.2 Environment configuration 9.3 Hosting of frontend and backend	68-73
10	Challenges & Limitations	10.1 Issues faced during development 10.2 Solutions applied 10.3 Current limitations or known bugs	74-80
11	Future Enhancements	11.1 Planned features 11.2 Possible integrations or optimizations	81-88

12	Conclusion	12.1 Summary of the project 12.2 What was achieved 12.3 Skills learned during development	89-100
13	References	- Books, tutorials, APIs, documentation sites used	
14	Appendices	- Screenshots of the app - Sample code snippets - Installation/setup instructions - User manual or guide -certifications	

CHAPTER 1

Introduction

1.1 Motivation

The increasing adoption of electric vehicles (EVs) has created a growing demand for charging stations and efficient charging facilities. EV users often face difficulties in finding nearby charging stations, understanding charging costs, and planning suitable routes. This motivated us to develop **EVision – Electric Vehicle Charging Demand Analytics**, a web-based application that helps users explore EV charging infrastructure and understand charging-demand information in Hyderabad through a single platform.

1.2 Objective

The main objective of EVision is to provide a user-friendly platform for exploring electric vehicle charging infrastructure and demand-related information in Hyderabad.

The system focuses on the following objectives:

- To display EV charging station locations.
- To provide an interactive charging-station map.
- To help users explore charging stations.
- To provide route-planning functionality.
- To estimate EV charging costs.
- To visualize electricity consumption and demand trends.
- To support future demand forecasting.

- To help understand charging infrastructure requirements.

1.3 What is Web Development?

Web development is the process of creating and maintaining websites and web applications that run through web browsers. It includes designing user interfaces, writing code, developing application features, and connecting different components. In our EVision project, React.js and TypeScript are used for frontend development, Vite is used as the development tool, and Tailwind CSS is used for designing the application interface.

1.4 What is Website?

A website is a collection of related web pages and resources that can be accessed through the Internet using web browsers such as Google Chrome and Microsoft Edge. Websites are developed for different purposes, including education, business, communication, and information sharing. EVision is a web-based application that provides EV charging station information, map visualization, route planning, cost estimation, and demand analytics.

1.5 What is Webpage?

A webpage is a single document or screen available within a website. It can contain text, images, buttons, tables, charts, and other interactive elements. In EVision, different screens are provided for the dashboard, charging-station discovery, station map, route planning, charging-cost estimation, demand analytics, forecasting, and infrastructure planning. These screens help users access the application's features in an organized manner.

CHAPTER 2

Software Requirement

2.1 Problem Statement

The increasing adoption of electric vehicles in Hyderabad has created a growing demand for charging stations and efficient charging facilities. EV users may face difficulties in finding suitable charging stations, understanding charging costs, and planning convenient routes. Similarly, infrastructure planners need information about charging-station locations and electricity-consumption patterns to identify areas that may require additional charging facilities.

Therefore, EVision is developed as a web-based application that brings together charging-station discovery, map visualization, route planning, charging-cost estimation, and demand analytics to make EV charging information easier to access and understand.

2.2 Technologies used

1. React.js and TypeScript (Frontend Development)

- **React.js:** React is a JavaScript library used to create interactive user interfaces. It helps develop reusable components for different application modules.
- **TypeScript:** TypeScript adds static typing to JavaScript, improving code readability and maintainability.

Both are used in EVision to develop the dashboard, navigation, charging-station pages, maps, and analytics interface.

2. Vite (Development and Build Tool)

Vite is a modern development and build tool used to run React applications. It provides a fast development server and helps prepare the application for deployment.

3. Tailwind CSS (Styling and Layout)

Tailwind CSS is a utility-first CSS framework used to design responsive web interfaces. It provides styling utilities for colors, spacing, layouts, buttons, and components. In EVision, it is used to create a clean and user-friendly interface.

4. Lucide Icons

Lucide is an icon library used to provide consistent icons for web applications. In EVision, icons improve navigation and help users identify different application features.

2.3 Software Used

2.3.1 Visual Studio Code

Visual Studio Code is a source-code editor used to develop and manage the project. It supports TypeScript, JavaScript, and CSS. It provides features such as an integrated terminal, extensions, debugging support, and code editing facilities.

2.3.2 Node.js and npm (Node Package Manager)

Node.js provides the JavaScript runtime environment required to run the frontend development tools. npm is used to install and manage the packages required for the React application, such as React, Vite, and other dependencies.

2.3.3 Modern Web Browsers (Google Chrome / Microsoft Edge)

Web browsers are used to access and run the EVision application. They display the user interface and allow users to interact with different modules. Google Chrome and Microsoft Edge can be used to check the application layout, navigation, maps, and other features.

CHAPTER 3

Software Design

3.1 DFD

3.1 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data moves through a system. It shows the input data, processes, output data, and interaction between users and the application.

In EVision, the DFD explains how users interact with the application, select different modules, and view EV charging-related information. The application processes locally available sample data and displays the results through dashboards, maps, cards, and visualizations.

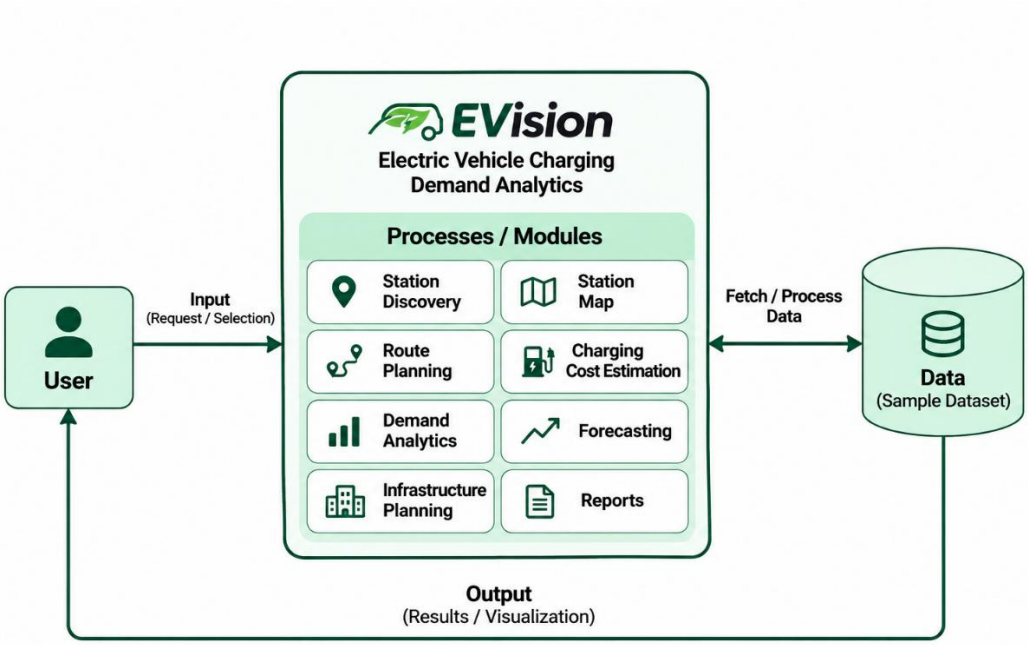


Fig No. 3.1

CHAPTER 4

Implementation and User Interface

4.1 Web Page Details

EVision consists of the following main modules:

1. Dashboard
2. Charging Station Discovery and Map
3. Route Planning and Cost Estimation
4. Demand Analytics
5. Demand Forecasting
6. Infrastructure Planning and Reports

4.2 Detailed Description of Pages

Dashboard

Displays an overview of EVision and provides access to its main features.

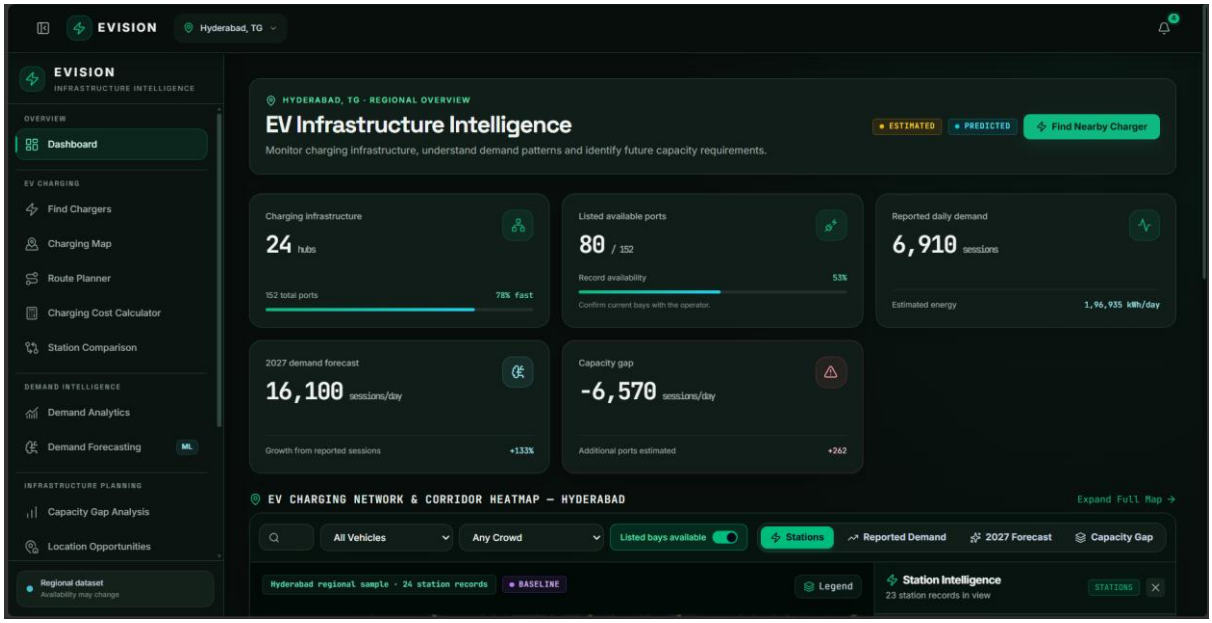


Fig No. 4.1 Dashboard Page

2. Charging Station Discovery and Map

Helps users explore charging stations and view their geographical locations.

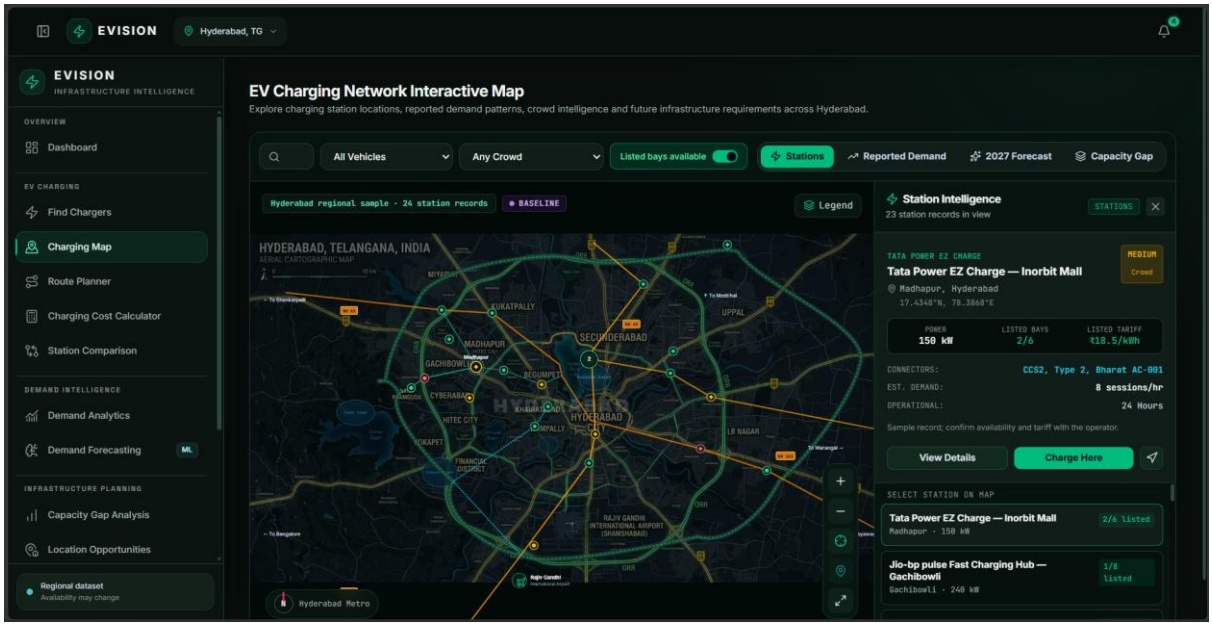


Fig No. 4.2 Charging Station Page

3. Route Planning and Cost Estimation

Provides route-planning features and estimated EV charging costs.

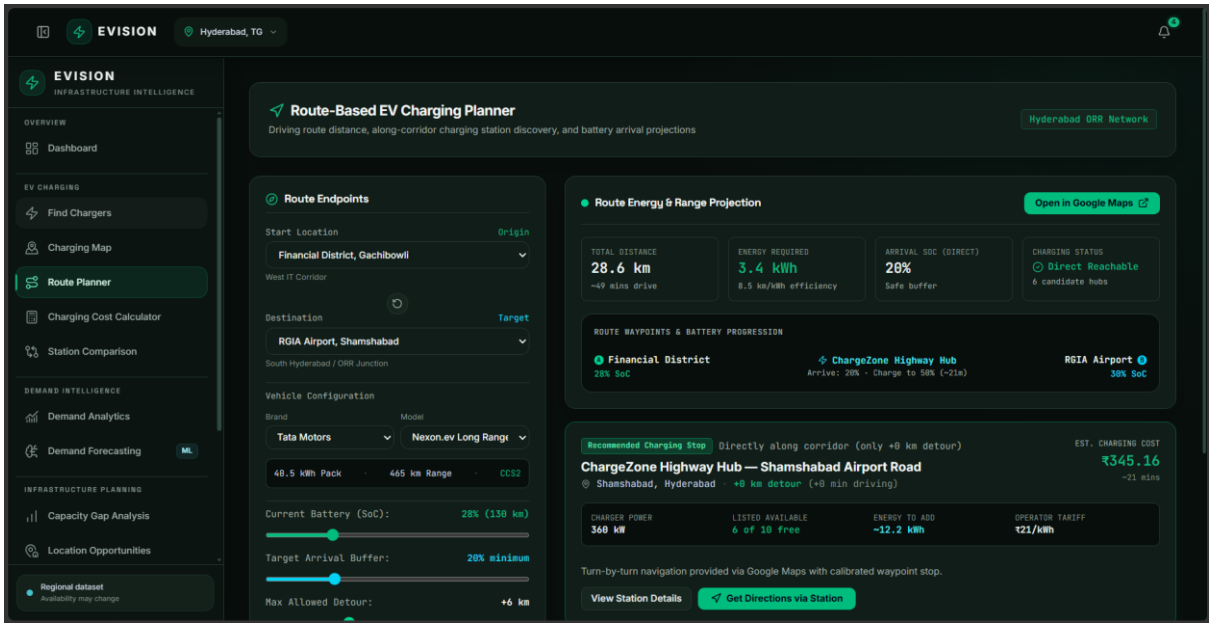


Fig No. 4.3 Route Planning and Cost Estimation Page

4. Demand Analytics

Displays charging-demand patterns through charts and visualizations.

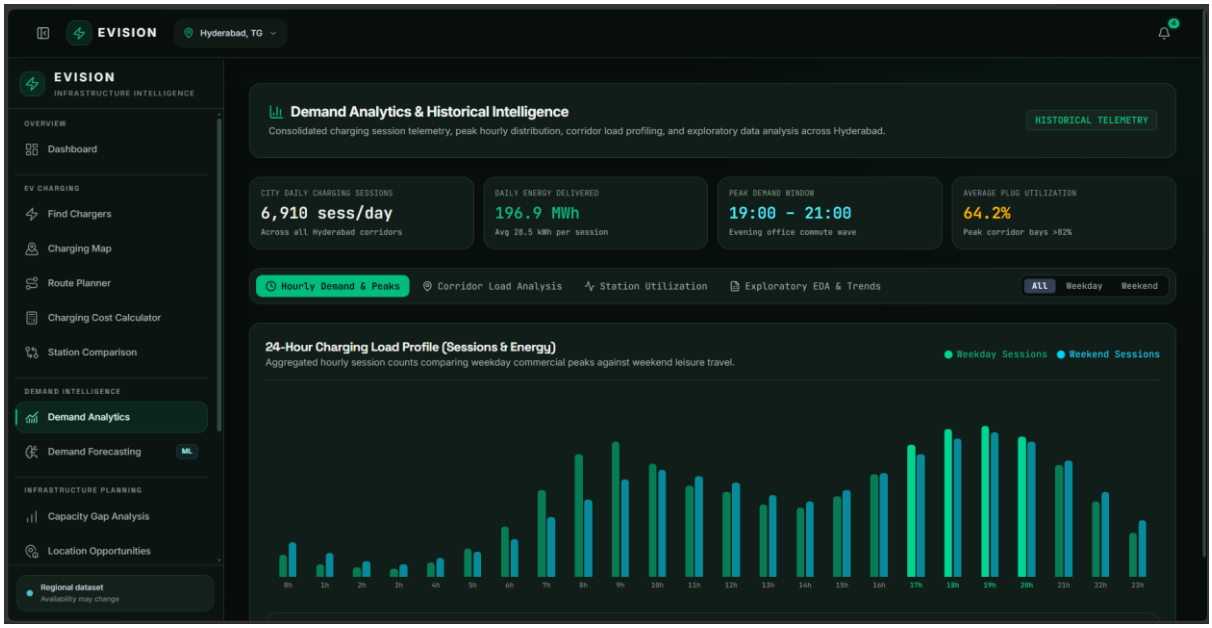


Fig No. 4.4 Demand Analytics Page

5. Demand Forecasting

Displays sample future EV charging-demand predictions.

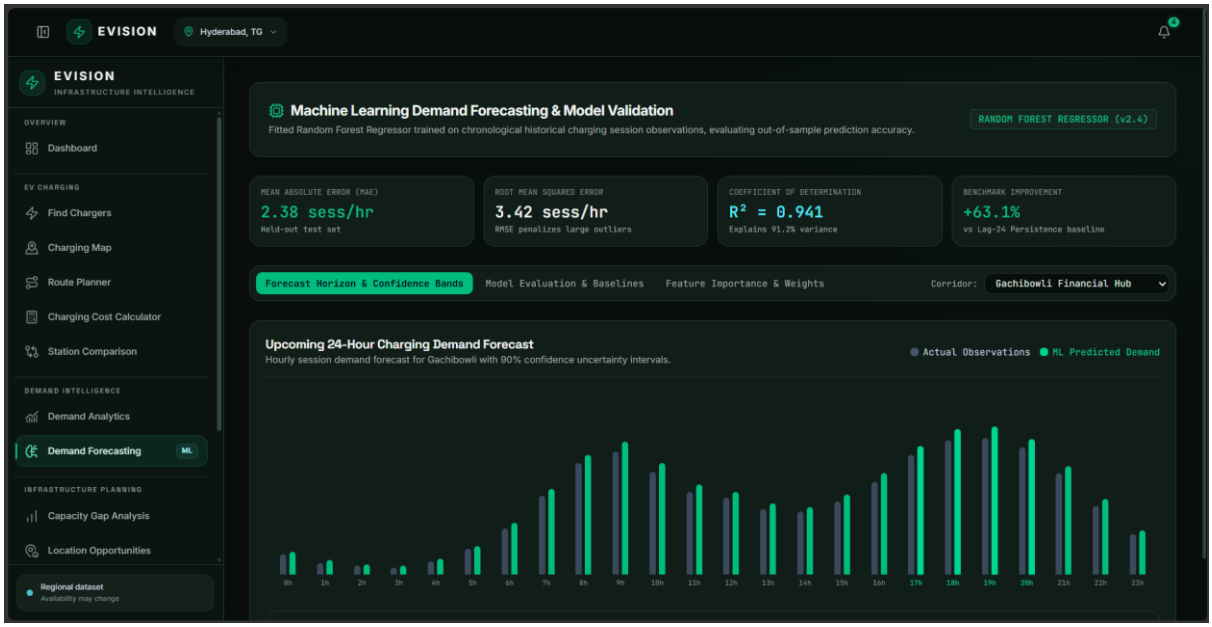


Fig No. 4.5 Forecasting Page

6. Infrastructure Planning and Reports

Presents infrastructure planning information and summarized reports.

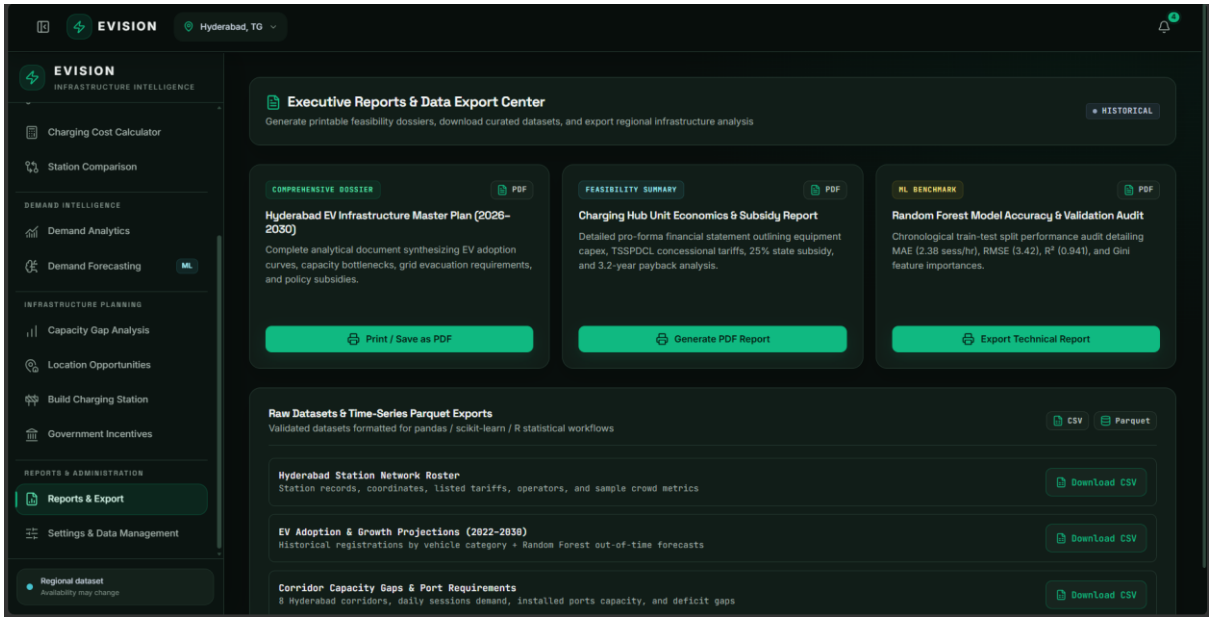


Fig No. 4.6 Infrastructure Planning and Reports Page

CHAPTER 5

Conclusion and Future Aspects

5.1 Conclusion

EVision – Electric Vehicle Charging Demand Analytics is a web-based application developed to support EV charging station discovery, route planning, cost estimation and demand analysis. It provides an interactive dashboard to visualize charging-related information and support better infrastructure planning.

5.2 Future Aspects

The project can be enhanced through the following features:

- Integration of real-time EV charging station data.
- Implementation of a machine-learning model for demand forecasting.
- Integration of a database for storing charging information.
- Improved route planning and charging cost estimation.
- Cloud deployment for wider accessibility.
- Enhanced security and user authentication.

CHAPTER 6

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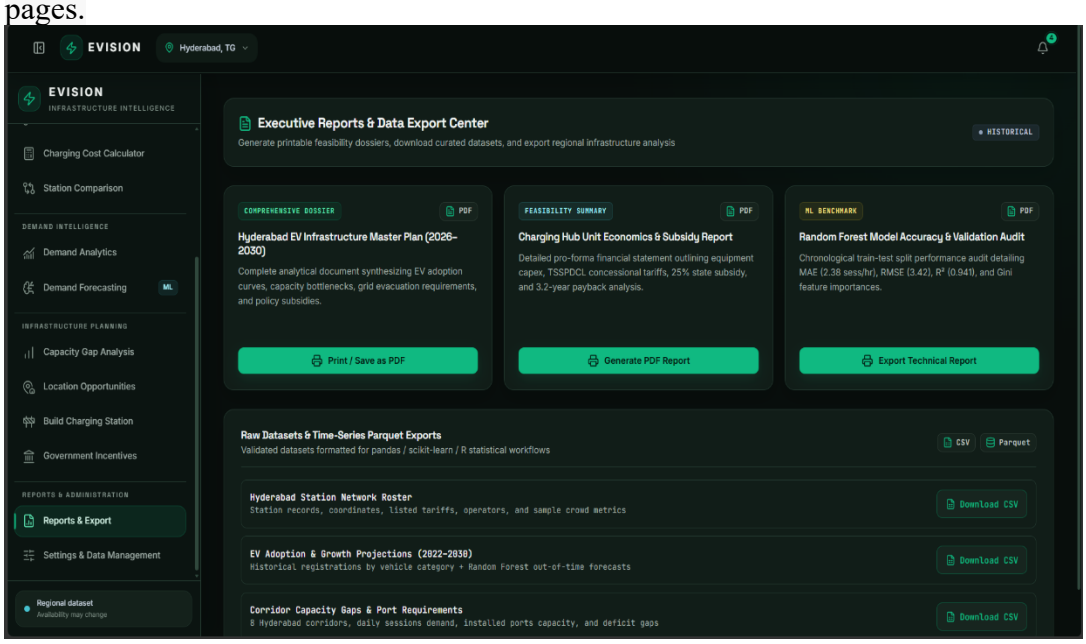
[6] K. Baek, E. Lee and J. Kim, "A Dataset for Multi-Faceted Analysis of Electric Vehicle Charging Transactions," *Scientific Data*, vol. 11, 2024.

CHAPTER 7

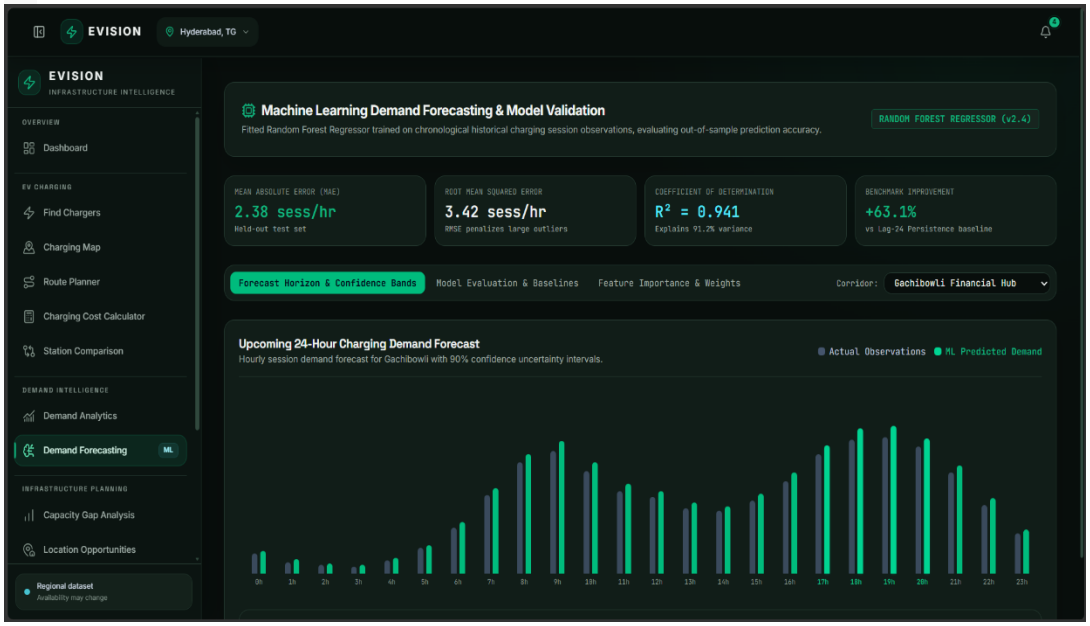
Appendices

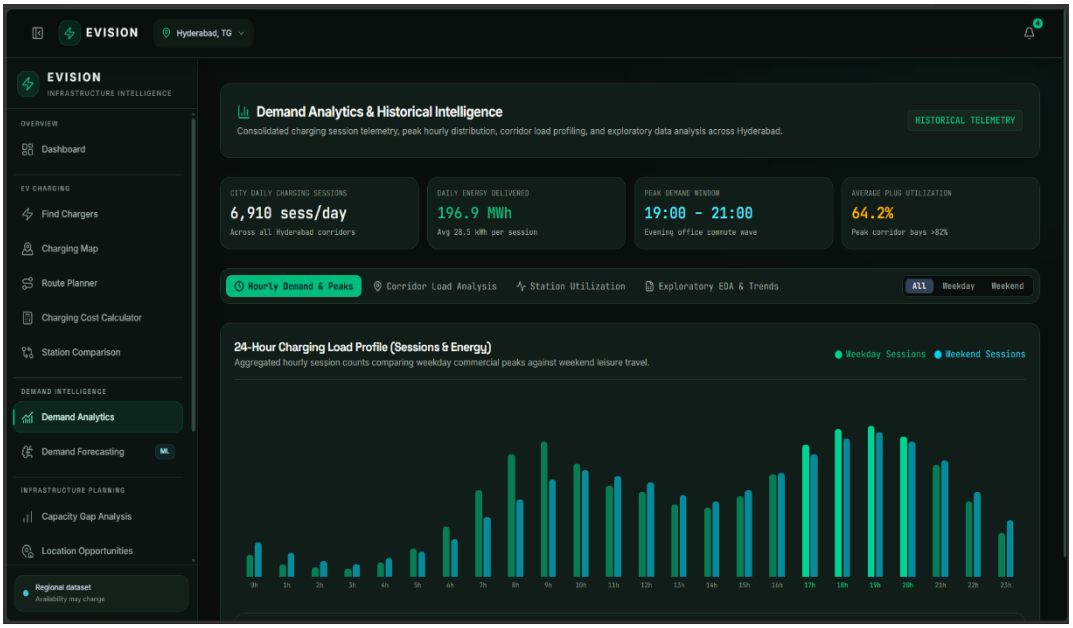
Appendix A – Application Screenshots

This section contains screenshots of the EVision application, including the dashboard, charging station map, route planning, demand analytics and forecasting pages.

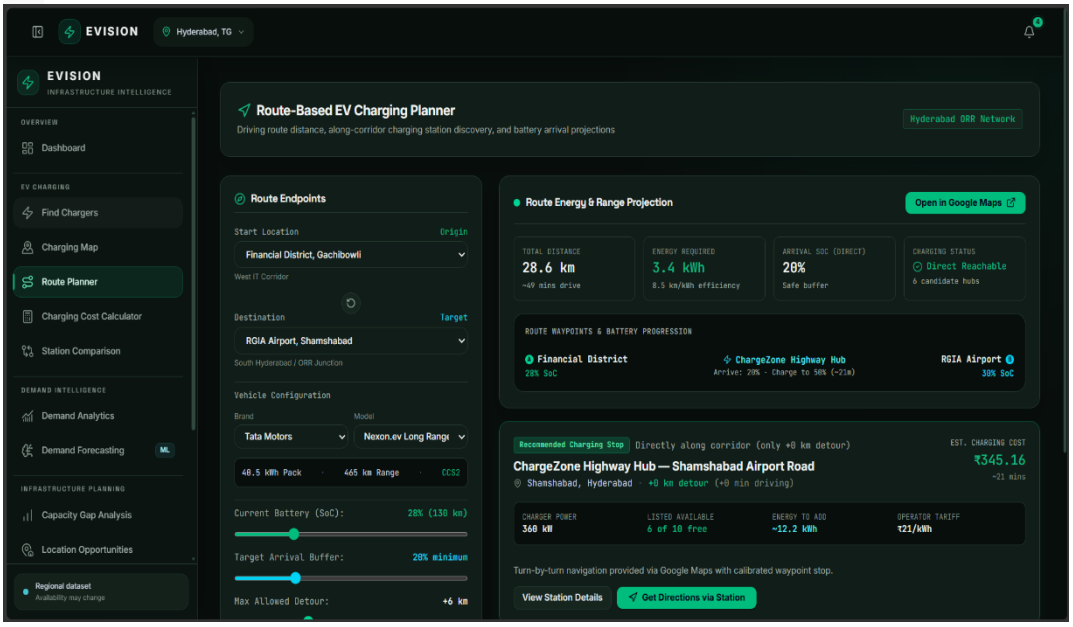


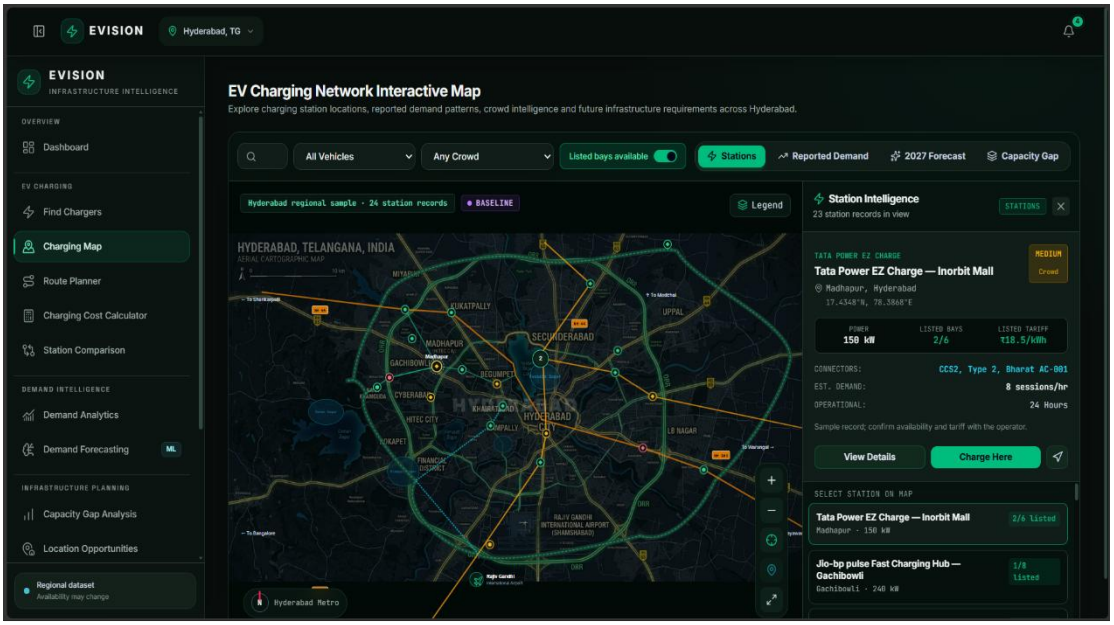
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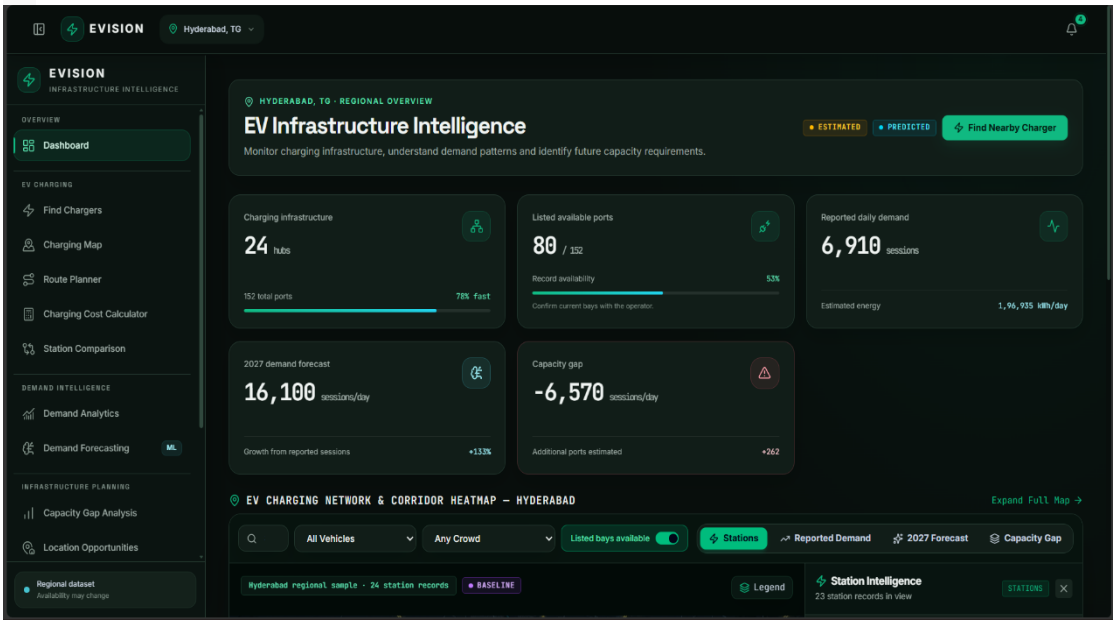

EVISION
Hyderabad, TG
EVISION
INFRASTRUCTURE INTELLIGENCE
OVERVIEW
Dashboard
EV CHARGING
Find Chargers
Charging Map
Route Planner
Charging Cost Calculator
Station Comparison
DEMAND INTELLIGENCE
Demand Analytics
Demand Forecasting
ML
INFRASTRUCTURE PLANNING
Capacity Gap Analysis
Location Opportunities
Regional dataset
Availability may change
Demand Analytics & Historical Intelligence
Consolidated charging session telemetry, peak hourly distribution, corridor load profiling, and exploratory data analysis across Hyderabad.
HISTORICAL TELEMETRY
CITY DAILY CHARGING SESSIONS
6,910 sess/day
Across all Hyderabad corridors
DAILY ENERGY DELIVERED
196.9 MWh
Avg 28.5 kWh per session
PEAK DEMAND WINDOW
19:00 - 21:00
Evening office commute wave
AVERAGE PLUG UTILIZATION
64.2%
Peak corridor bays >82%
Hourly Demand & Peaks
Corridor Load Analysis
Station Utilization
Exploratory EDA & Trends
All
Weekday
Weekend
24-Hour Charging Load Profile (Sessions & Energy)
Aggregated hourly session counts comparing weekday commercial peaks against weekend leisure travel.
Weekday Sessions
Weekend Sessions
0h
1h
2h
3h
4h
5h
6h
7h
8h
9h
10h
11h
12h
13h
14h
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Demand Forecasting
ML
INFRASTRUCTURE PLANNING
Capacity Gap Analysis
Location Opportunities
Regional dataset
Availability may change
Route-Based EV Charging Planner
Driving route distance, along-corridor charging station discovery, and battery arrival projections
Hyderabad ORR Network
Route Endpoints
Start Location
Origin
Financial District, Gachibowli
West IT Corridor
Destination
Target
RGIA Airport, Shamshabad
South Hyderabad / ORR Junction
Vehicle Configuration
Brand
Model
Tata Motors
Nexon.ev Long Range
40.5 kWh Pack
465 km Range
CCS2
Current Battery (SoC):
28% (130 km)
Target Arrival Buffer:
20% minimum
Max Allowed Detour:
+6 km
Route Energy & Range Projection
Open in Google Maps
TOTAL DISTANCE
28.6 km
~49 mins drive
ENERGY REQUIRED
3.4 kWh
8.5 km/kWh efficiency
ARRIVAL SOC (DIRECT)
20%
Safe buffer
CHARGING STATUS
Direct Reachable
6 candidate hubs
ROUTE WAYPOINTS & BATTERY PROGRESSION
Financial District
28% SoC
ChargeZone Highway Hub
Arrive: 20% · Charge to 50% (~21m)
RGIA Airport
50% SoC
Recommended Charging Stop
Directly along corridor (only +0 km detour)
EST. CHARGING COST
ChargeZone Highway Hub — Shamshabad Airport Road
₹345.16
Shamshabad, Hyderabad · +0 km detour (+0 min driving)
~21 mins
CHARGER POWER
360 kW
LISTED AVAILABLE
6 of 10 free
ENERGY TO ADD
~12.2 kWh
OPERATOR TARIFF
₹21/kWh
Turn-by-turn navigation provided via Google Maps with calibrated waypoint stop.
View Station Details
Get Directions via Station



...



Appendix B – Technologies Used

- Frontend: React.js
- Programming Language: TypeScript
- Build Tool: Vite
- Styling: Tailwind CSS

- Icons: Lucide React

Appendix C – Application Execution

The following commands are used to run the application locally:

1. Install dependencies: npm install
2. Start the application: npm run dev
3. Build the application: npm run build

Appendix D – Project Source Code

The source code contains the frontend components, application pages, styling files and sample data used in EVision.

